

JAVA Table of Content- 28th Jan'26

Day	Topic	Updated Teaching Session
Day 1	PLT	PLT
Day 2	1) Programming Basics i) Various basic java programming Concepts, operators ii) Conditional statement, loops, switch cases iii) Java Array Manipulation	Introduction Teaching Session - 1 Various basic java programming Concepts, Conditional statement, switch cases, Teaching session - 2 Various types of loops Teaching session - 3 Array Part-1
Day 3	2) Arrays and strings i) Various methods related to string	Doubt clearing session Teaching Session - 1 Arrays Part-2 Teaching session - 2 String Teaching session - 3 String Methods
Day 4	3) OOPS programming i) OOP basics ii) Method overloading Static Members & Methods iv) Inheritance iii)	Doubt clearing session Teaching Session-1 OOPS Teaching Session-2 Method Overloading, Static Members & methods Teaching Session-3 Inheritance
Day 5	4) Run time polymorphism Final variable & methods, Final class ii) Abstract class & Methods iii) Method Overriding iv) Interface i)	Doubt clearing session Teaching Session-1 Method Overriding Teaching Session-2 Abstract Class & Methods Teaching Session-3 Interface
Day 6	5) Exception Handling i) try catch finally ii) throw, throws iii) custom exceptions	Doubt clearing session Teaching Session-1 try catch Teaching Session-2 throw & throws Teaching Session-3 custom exceptions (Weekly Assessment -1)
Day 7	6) Collection ArrayList ii) LinkedList iii) HashSet iv) TreeSet v) Comparable & Comparator i)	Doubt clearing session Teaching Session-1 Teaching Session-2 HashSet & TreeSet Teaching Session-3 Comparable & Comparator
Day 8	7) Map HashMap ii) Tree Map iii) LinkedHashMap i)	Doubt clearing session Teaching Session-1 HashMap Teaching Session-2 LinkedHashMap Teaching Session-3 Tree map
Day 9	Milestone Preparation (Extra Day 1)	Milestone Preparation (Extra Day 1)
Day 10	8) RDBMS Basics & MySQL Queries i) RDBMS Basics, Select, update, delete, insert queries and Joins ii) JDBC Basics	Doubt clearing session Teaching Session-1 RDBMS Basics

		Teaching Session-2 1	JDBC
		Teaching Session-3 2	JDBC
Day 11	JDBC Crud Operations	Doubt clearing session	
		Teaching Session-1 Results Set	
		Teaching Session-2	Maven
		Teaching Session-3	Html, CSS
Day 12	Milestone Preparation + Mock Assessment 1		
Day 13	Milestone Preparation (Extra Day 2)		
Day 14	Milestone Preparation + Mock Assessment 2		
Day 15	Milestone 1		
Day 16	6) Webservices	Doubt clearing session	
	i) Introduction	Teaching Session - 1	
	ii) Client and server communications	Understanding of client and server technology	
	iii) What is a Tomcat server	Webservices Introduction	
	iv) What is a servlet	Webserver (Example Tomcat)	
	v) What http protocol	Teaching session - 2	
	vi) JSON and xml	Understanding of servlet .	
		How Servlet works.	
		Teaching session - 3	
		Understanding of HTTP protocol , methods	
		What is JSON + PRE CEFR	
		Experiential Learning / Incremental Project Learning -Day1	
Day 17	7) Spring framework	Doubt clearing session	
	i) Spring IOC Introduction	Teaching Session-1	
	ii) Spring DI and Autowiring	DI example and IOC container	
	iii) Introduction to Spring framework	- How Runtime polymorphism concept behind DI	
	iv) Hello world program in Springboot	Teaching Session-2	
		Autowiring concept and annotations	
		Introducing Springboot	
		Teaching Session-3	
		Introducing Springboot	
		Experiential Learning / Incremental Project Learning -Day2	
Day 18	7) spring framework [Continued] And 8) Springboot	Doubt clearing session	
	i) Various layers in Rest API	Teaching Session-1	
	ii) Ability to implement Exception Handling,	Springboot Get mapping and Post mapping	
	iii) Validation	Using Postman	
	iv) Various Response Code	Necessary annotations	
	v) CRUD operation with collections.	Teaching Session - 2	
		Various layer in Springboot REST API.	
		Various Response codes(404, 200 ,500 etc)	
		Teaching session - 3	
		Use of Exception handling on Springboot.	
		Creation of Custom Exception on Springboot+Milestone 1 Retest	
		Experiential Learning / Incremental Project Learning -Day3	
Day 19	9) ORM Tool	Doubt clearing session	
	i) What is ORM tool.	Teaching session - 1	
	ii) Diff between ORM and JDBC	Springboot application for CRUD operations using Collections.	
	iii) Hibernate and JPA example	Teaching session - 2	
	iv) Configurations and Annotations	ORM Introduction.	
	v) Crud operations with JPA /Hibernate	Advantage of ORM over JDBC	
		Teaching Session-3	
		Use of Hibernate/JPA	
		Stereotype Annotations	
		CRUD operations using necessary annotations	
		Experiential Learning / Incremental Project Learning -Day4	

Day 20	10) Spring data JPA i) Spring Boot JPA introduction ii) Single entity CRUD by Springboot JPA iii) JPQL (eg : find by name query)	Doubt clearing session Teaching Session-1 Various JPA Annotations Teaching Session-2 Introduction of Springboot JPA. Teaching Session-3 JPQL Experiential Learning / Incremental Project Learning -Day5
Day 21	11) Normalization and RDBMS i) What is Normalization ii) Referential integrity constraints iii) Normalization up to 3NF iv) Implementation of Normalization based on project related entities With help of problem statement .	Doubt clearing session Teaching Session - 1 Normalization Teaching session - 2 Recap Teaching session - 3 Weekly Assessment 2 Experiential Learning / Incremental Project Learning -Day6
Day 22	12) Springboot JPA Mapping i) Various mapping in Spring Boot ii) Associations and appropriate annotations	Doubt clearing session Teaching Session-1 One to One association and crud operation using Springboot JPA. Teaching Session-2 One to Many association and crud operation using Springboot JPA. Understanding of Mapped by. Teaching Session-3 Many to One association and crud operation using Springboot JPA with One more example Experiential Learning / Incremental Project Learning -Day7
Day 23	12) Springboot JPA Mapping [Continued] i) Many to many and many to many example ii) Commonly used / Necessary Annotations ii) Debugging of Spring boot application	Doubt clearing session Teaching Session-1 Many to Many association and crud operation using Springboot JPA Teaching Session-2 Debugging of Spring boot application and bug fixing , other Guidelines Teaching Session-3 Debugging of Spring boot application and bug fixing , other Guidelines Experiential Learning / Incremental Project Learning --Day8
Day 24	Spring Security i) Spring Boot application with basic security controls ii) Authentication, Authorization iii) Securing endpoints with HTTP Basic authentication	Doubt clearing session Teaching session - 1 Understanding token based Spring boot security. Teaching Session - 2 Implementing JWT token Teaching Session - 3 Implementing JWT Token ,GIT Upload Experiential Learning / Incremental Project Learning -Day9
Day 25	Spring Security[Continued] i) JWT Token	Doubt clearing session Teaching session - 1 Implementing JWT Token Teaching session - Implementing JWT Token Teaching Session-3 Implementing JWT Token , GIT Upload Experiential Learning / Incremental Project Learning -Day10
Day 26	Version control System	Doubt clearing session

	Git and GitHub introduction	Teaching Session-1 What is Version control and why it is required. GIT introduction and GIT CLI Teaching Session-2 Repository management : GIT various commands like GIT Clone , commit , push etc. Teaching Session-3 Repository management : GIT various commands like GIT Clone , commit , push etc. , Git Upload Experiential Learning / Incremental Project Learning - -Day11
Day 27	Version control System [Continued] i)Integrate IDE with Git - VS and VS code ii)Git basics and CLI iii)Repositories management iv)Git files, cloning, check-in and commit v)Git branch and merge conflicts vi)GitHub vii)Fork and push, pull, sync	Doubt clearing session Teaching Session-1 Branch creation on GIT HUB and push code to new branch. Teaching Session - 2 Merge conflict Teaching Session-3 Understanding of Spring security basic. Understanding of Authentication and Authorization, Git upload Experiential Learning / Incremental Project Learning -Day 12
Day 28	Milestone 2	Doubt clearing session Teaching Session-1 Recap Teaching Session-2 Recap Milestone 2
Day 29	15) JavaScript i) Basics , Iterators ii) Classes, methods iii)Functions	Doubt clearing session Teaching session - 1 Understanding of JavaScript Learn and apply both inline JavaScript , reference JavaScript. Teaching Session - 2 Java script object and array creation. JavaScript Basics , Constructs Classes and methods Teaching session - 3 JavaScript functions and return types. Arrow Functions Experiential Learning / Incremental Project Learning - Day 13
Day 30	15) JavaScript [Continued] ii)Create events and DOM manipulations. iii)Able to debug JavaScript	Doubt clearing session Teaching Session - 1 JavaScript Class , Properties and objects . Teaching session - 2 JavaScript Class , Properties and objects . Teaching session - 3 Java script events and Dom manipulations. Experiential Learning / Incremental Project Learning - Day 14
Day 31	16) Typescript i) what is Type script ii) building blocks iii) Strings iv) Arrays v) functions vi) Java Script ES6 vii)Arrow Functions viii) Promises	Doubt clearing session Teaching Session-1 Typescript Datatypes , Strings manipulations Functions Teaching Session-2 Arrays and Various manipulations Advanced Array. Teaching Session-3 What is Promises Advantage and usage of Promises +Milestone 2 Retest Experiential Learning / Incremental Project Learning -Day 15
Day 32	17) Angular i) Installation and helloworld program ii)Understanding of Angular various components. lii)Data binding -interpolation	Doubt clearing session Teaching Session - 1 Angular Basics Recap

		Teaching session - 2 Various packages and files. Angular CLI Usage Using Components Template and Template URL Teaching session - 3 Data binding Interpolation Experiential Learning / Incremental Project Learning -Day 16
Day 33	17) Angular [Continued] ii) Various types of data binding - data binding, two way binding , property binding , event binding	Doubt clearing session Teaching Session-1 One way binding Property Binding Teaching Session-2 Event binding and why it is required. Teaching Session-3 Two way Binding , Use of NgModel Experiential Learning / Incremental Project Learning -Day 17
Day 34	17) Angular [Continued] i) Inter component communication.	Doubt clearing session Teaching Session-1 Router Parameters Query Params Teaching Session-2 @Input and @Output Teaching Session-3 Experiential Learning / Incremental Project Learning - Day 18
Day 35	17) Angular [Continued] i) Directives ii) Pipes	Doubt clearing session Teaching Session - 1 What is Directives and why it is required. Types of Directives . Teaching session - 2 Various directives like ngIf , ngClass ngFor etc. Teaching session - 3 Use PIPES to transform data for display. Weekly Assessment 3 Experiential Learning / Incremental Project Learning - Day 19
Day 36	17) Angular [Continued] i) Routing ii) Forms and validation iii) Template Driven form iv) Reactive form, Validators	Doubt clearing session Teaching Session - 1 Routing : Routing configuration and use of Routing to create header , footer and navigation. Teaching Session - 2 Using Template Driven form for passing data between forms. Reactive form: Use of reactive form to dynamically display form data. Teaching Session - 3 Form validation using angular Validators Experiential Learning / Incremental Project Learning - - Day 20
Day 37	17) Angular [Continued] i) Services ii) Observables iii) HttpClient	Doubt clearing session Teaching Session-1 Use of Services and Observable Retrieve hardcoded data from any JSON or Array object by using Services. Teaching Session-2 Use of Services and Observable [Continued] Teaching Session-3 Use of Http client for retrieving data from Springboot Experiential Learning / Incremental Project Learning - Day 21
Day 38	17) Angular [Continued] i) Angular Auth	Doubt clearing session Teaching Session - 1 HttpClient Recap, Integration of Back End to Front End

		Teaching session - 2 Angular Auth : Use of Angular auth for Admin and user Login / dashboard. Use of Angular Auth guard Teaching session - 3 Angular Auth : [Continued] Experiential Learning / Incremental Project Learning -Day 22
Day 39	17) Angular [Continued] i) Bootstrap ii) Jasmine and Karma	Doubt clearing session Teaching Session-1 Implementation and uses of Bootstrap components Teaching Session - 2 Implementation of Jasmine and Karma Teaching session - 3 Implementation of Jasmine and Karma [continued] Experiential Learning / Incremental Project Learning -Day 23
Day 40	18) JUnit i) What is test and test case ii) What is unit testing iii) Introduction about JUnit iv) Writing Various Unit test cases 19) Agile i) Agile methodology ii) Necessary methods and guidelines for upcoming project gladiator phase	Doubt clearing session Teaching Session -1 JUnit and why it is required Teaching Session - 2 JUnit common Annotations used in unit testing, Mockito. Mocking of the data for unit testing. Teaching Session - 3 Agile ,Scrum , Sprint ,Agile methodology to be used in project Gladiator Experiential Learning / Incremental Project Learning -Day 24
Day 41	Recap of the End to end application Any Left Over Topics Revision Mock Que Ans practices doubt solving etc	Recap , Experiential Learning project / Incremental project enhancements
Day 42	Recap of the End to end application Any Left Over Topics Revision Mock Que Ans practices doubt solving etc .	Recap , Experiential Learning project / Incremental project enhancements
Day 43	Milestone 3	Milestone 3
Day 44	Mock Interview & Incremental Project Evaluation	Mock Interview & Incremental Project Evaluation
Day 45	Case Study Implementation by associates in Group	Case Study Implementation by associates in Group 1. Daily SME Connect Expectation settings , Providing guideliens , monitoring progress, resolving doubts with flow of project. 2. Checking Group Progress on platfrom Day 5 - 50% test case passing is expected Day 10 - 100% test case passing is expected
Day 46		Milestone 3 Retest
Day 47		
Day 48		
Day 49	Case Study Implementation by associates in Group	3. Currently Trainer will maintain excel sheet for daily progress of each learner One Day before final presentation day - review against each learners will be provided by Trainer . 4. Vendor should be able to provide the daily progress of each learner as an autoevalated report
Day 50		
Day 51		
Day 52		
Day 53		
Day 54		
Day 55		
Day 56	Project Evaluation	Project Evaluation